Previous Work Review & Research Directions

Energy-Efficient LLM Serving and HBM/HBF Co-Design

Research meeting: 4 November 2026

Yu-Jui Liang Advisor: Hsiang-Yun Cheng
National Taiwan University / Academia Sinica

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Power-, Energy-, and Carbon-Aware GPU Serving Optimization

How can conventional GPU serving systems optimize energy, carbon, and resource allocation?

GreenCache; PRISM / Beyond Energy

Cache Your Prompt When It's Green — Carbon-Aware Caching for Large Language Model Serving

Motivation

- **Caching trade-off:** Reusing SSD-based prefix KV caches avoids prefill computation and operational carbon from electricity, but adds embodied carbon from SSD manufacturing.
- **Variable benefits:** Carbon-efficient cache capacity changes with request load and grid carbon intensity. Larger caches are not always greener.
- **Outcome:** Llama-3 70B in France: 15.1% average carbon reduction vs. Full Cache, peaking at 25.3% in multi-turn serving. Reported latency SLO attainment: $> 90\%$.

Method

- **Profile:** Measure latency and power across cache capacities and request loads. Estimate operational and allocated embodied carbon, capturing KV-reuse benefits.
- **Allocate:** Predict load and grid carbon intensity. An ILP selects provisioned SSD cache capacity to minimize carbon subject to TTFT/TPOT SLO-attainment constraints.
- **Replace:** Least Carbon Savings (LCS) scores KV entries using reused tokens, hit frequency, size and age, evicting entries with the lowest carbon-benefit scores.

Beyond Energy: When Sustainability Dimensions Reshape LLM Serving Decisions

Motivation

- **Sustainability trade-off:** Minimizing IT energy need not minimize carbon, water or biodiversity impacts across deployment regions and times.
- **Scope matters:** For a fixed workload and region/time, all four operational-only rankings follow configuration energy. Deployment or embodied impacts can change rankings.
- **Outcome:** 50.2% median reduction in worst-case regret vs. offline WaterWise in the 12-region evaluation (20 capacity seeds).

Method

- **Model:** Combine IT energy with regional environmental intensities to evaluate four operational impacts. Add embodied impacts for lifecycle comparisons.
- **Compare:** Separate GPU/model configuration rankings from region/time deployment rankings. Identify when operational rankings agree and embodied impacts reverse them.
- **Route:** Allocate workloads across regions to minimize the largest relative regret vs. each dimension's optimum, subject to regional capacity and serving constraints.

HBF System Characterization and KV-Centric Serving

When does HBF improve LLM serving, and when does it become a bottleneck?

Characterizing HBF; HBF Sucks?; Hot–Cold Tiering

Characterizing High Bandwidth Flash for LLM Serving

Motivation

- **Capacity pressure:** Long contexts and agentic calls compete for HBM space for weights, active KV and reusable prefixes.
- **Retention matters:** More HBF capacity helps only when placement and scheduling account for read costs, slow writes and finite endurance.
- **Outcome:** Trace-driven simulation improves completion time and estimated write lifetime. Modeled energy savings exclude static power.

Method

- **Organize:** Model an HBM–HBF–host hierarchy and compare weight placement and KV residency under different memory organizations.
- **Retain:** Track prefix reuse and cache movement. Buffer growing KV in HBM and preserve useful prefixes to avoid reloads or recomputation.
- **Schedule:** Prioritize cache hits and limit admission with buffered cache-aware scheduling, balancing batch growth against future KV reuse.

HBF Sucks? A Full-Stack Characterization of High-Bandwidth Flash for KV-Centric LLM Serving

Motivation

- **Drop-in risk:** Faster flash does not ensure faster serving when HBF replaces SSD storage but also consumes near-tier package resources.
- **Traffic mismatch:** Hot KV remains in the near tier, leaving HBF a cold, write-heavy stream with limited useful reads to repay writes.
- **Outcome:** The modeled Mooncake-style substitution can worsen latency and goodput. This is a conditional deployment result, not a rejection of HBF.

Method

- **Replay:** Extend TokenSim with GPU/near-tier/HBF costs and replay production KV-offload traces using capacity-matched SSD baselines.
- **Diagnose:** Test three conditions: reads on the critical path, enough useful reads per write, and bandwidth that can be sustained.
- **Constrain:** Combine thermal throttling and write-endurance models to distinguish peak device bandwidth from practical serving benefits.

Hot–Cold Tiering of HBM and High Bandwidth Flash for Agentic LLM Serving

Motivation

- **Idle state:** Agentic sessions pause for tools or users while retaining large KV histories that exceed HBM capacity.
- **Two access patterns:** Active KV is read every decode step; paused KV is read on resume. Serving all KV from HBF wastes read energy.
- **Outcome:** Trace-based evaluation expands retained-session capacity while keeping active decoding in HBM and reducing flash read power.

Method

- **Classify:** Use session activity to separate a small hot KV working set from a large cold pool of paused-session state.
- **Place:** Keep active KV in HBM. Evict cold KV to HBF under capacity pressure; immutable blocks with valid HBF copies need no rewrite.
- **Resume:** Prefetch paused KV back into HBM through buffered inter-tier transfers before execution resumes, overlapping movement with decode.

HBM/HBF Architecture and Data-Path Co-Design

How should GPU, HBM, and HBF be organized to improve memory efficiency?

H3; FlashAccel; HBFlex; Beyond Capacity

H3: Hybrid Architecture Using High Bandwidth Memory and High Bandwidth Flash for Cost-Efficient LLM Inference

Motivation

- **Capacity and cost:** Long-sequence inference needs more memory than HBM economically provides, while HBF trades capacity for access and write costs.
- **Access asymmetry:** Large read-only data can exploit HBF capacity; mutable data benefits from HBM latency and write endurance.
- **Outcome:** The reported simulations improve throughput per power over HBM-only systems, especially for large shared precomputed KV workloads.

Method

- **Partition:** Classify inference data by whether it is read-only or requires updates, rather than placing all state in a single medium.
- **Combine:** Use a hybrid HBM/HBF organization: place large read-only state in HBF and retain other data in HBM.
- **Exploit capacity:** Keep more requests resident with the same GPU count, evaluating the concurrency benefit against HBF access and power costs.

FlashAccel: Leveraging High-Bandwidth Flash (HBF) for High-Throughput LLM Inference

Motivation

- **Capacity wall:** Large model weights and KV histories restrict batch size and prefix retention in HBM-based GPUs.
- **Utilization gap:** HBF capacity alone cannot hide NAND latency or keep its many planes busy under changing inference accesses.
- **Outcome:** The simulated hardware/software design improves throughput and energy efficiency over HBM-only serving under a latency constraint.

Method

- **Hide latency:** Integrate GPU, HBM and HBF with distributed SRAM buffers that prefetch data ahead of GPU consumption.
- **Arrange data:** Lay out weights in access order and balance KV data across HBF planes to improve parallel reads and manage flash erasure.
- **Manage:** Expose heterogeneous resources through an HBF-aware storage layer and programming model, coordinating persistent data and execution.

HBFlex: A Flexible Memory System for Bridging Fine-Grained LLM States and Coarse-Grained HBF Parallel Execution

Motivation

- **Package trade-off:** Full-HBF designs devote more package resources to capacity and parallel bandwidth than hybrid designs retaining HBM.
- **Dynamic-state cost:** Fine-grained KV reads imbalance planes; incremental writes block reads; mixed KV lifetimes amplify garbage collection.
- **Outcome:** Trace-driven simulations improve throughput over the evaluated H3 and FlashAccel baselines through coordinated KV management.

Method

- **Balance reads:** Distribute KV placement and attention accesses across planes, exposing parallelism and reducing overloaded-plane delays.
- **Schedule writes:** Aggregate incremental KV updates and write them back during sufficiently long compute windows to reduce read interference.
- **Reclaim carefully:** Pack KV with similar lifetimes into blocks and defer reclamation, reducing valid-page migration and extra flash writes.

Beyond Capacity: Scalable MoE LLM Inference via High-Bandwidth Flash with Direct GPU and HBM Paths

Motivation

- **Expert capacity:** MoE expert weights outgrow HBM even though each token uses only a small subset of experts.
- **Delivery bottleneck:** A single HBM relay path can leave direct GPU–HBF bandwidth unused, while late expert selection exposes NAND latency.
- **Outcome:** Event-driven serving simulation shows higher expert-delivery efficiency and throughput than either delivery route alone.

Method

- **Use two paths:** DASH assigns whole experts to direct GPU–HBF or HBF–HBM-base-die–GPU routes and transfers them concurrently.
- **Look ahead:** Determine upcoming experts early enough to overlap HBF reads with preceding computation, reducing exposed read latency.
- **Separate traffic:** Isolate immutable expert weights from mutable KV; buffer small decode writes and coordinate larger prefill transfers.

HBF Simulation and Power/Thermal Modeling

How can HBF architectures be evaluated before physical hardware becomes available?

HBFSim (real GPU); HBFSim (event-driven); HBM Power

HBFSim: Fast and Faithful Simulation of High-Bandwidth Flash Under Real GPU Execution

Motivation

- **Evaluation gap:** Physical HBF is unavailable for validation; cycle-level GPU simulation is slow and fixed traces miss runtime changes.
- **Core idea:** Run the real workload on a GPU and model only the program-visible HBF memory behavior.
- **Outcome:** The paper demonstrates timing injection and logical-capacity experiments. Injection accuracy is not future-HBF prediction accuracy.

Method

- **Intercept:** Register modeled address ranges and instrument supported PTX memory accesses while real GPU kernels execute.
- **Model service:** Timing-only mode retains GPU data; Capacity mode uses backing storage and GPU frames. MQSim supplies media timing.
- **Check scope:** Public defaults use issue-point waits. Dependency waits are experimental; thermal/reliability modeling is optional and uncalibrated.

HBFSim: A Simulator for Exploring HBF in Memory Hierarchies for LLMs

Motivation

- **Design uncertainty:** HBM/HBF capacity, NAND latency and finite endurance make hierarchy choices difficult before hardware is available.
- **Flash semantics:** Placement must account for page programming, out-of-place updates, garbage collection and data movement.
- **Outcome:** The event-driven study finds active KV retention in HBM preferable to using HBM merely as an HBF cache in its evaluated workloads.

Method

- **Drive workloads:** Use a calibrated LLM serving frontend to generate weight, KV and scratch accesses with application dependencies.
- **Simulate hierarchy:** Model HBM, HBF and host tiers with event-driven resource calendars, links and host-managed flash translation.
- **Explore policies:** Vary placement and transfers; track NAND reads, programs, erases and write amplification to estimate serving time and endurance.

Understanding the Power Consumption of Modern High Bandwidth Memory: Experimental Analysis and Modeling Using Real HBM2 DRAM Chips

Motivation

- **Power uncertainty:** A fixed HBM power assumption misses differences across commands, banks and transferred data patterns.
- **Measured variation:** Real HBM2 measurements show that internal structure and bus toggling materially change memory power.
- **Outcome:** An empirical HBM model predicts measured power more accurately than evaluated prior models. This work characterizes HBM, not HBF.

Method

- **Measure:** Run controlled command and data-pattern experiments on 36 real HBM2 chips, separating structural and data-dependent effects.
- **Model:** Extend DRAMPower using measured currents and switching behavior to account for bank organization and data transitions.
- **Validate:** Compare predicted power with measurements and workload traces, including LLM decoding; separately check HBM3E GPU power.

HBF-Accelerated Retrieval and RAG Systems

Can HBF accelerate retrieval workloads beyond LLM decoding?

HAVEN: vector search and reranking

HAVEN: High-Bandwidth Flash Augmented Vector Engine for Large-Scale Approximate Nearest-Neighbor Search Acceleration

Motivation

- **Retrieval bottleneck:** Billion-scale RAG vector databases exceed GPU HBM. High-recall reranking must fetch full-precision vectors from DRAM or SSD.
- **Locality opportunity:** On-package HBF can retain large read-mostly vector data near the GPU and reduce off-GPU transfers.
- **Outcome:** Modeled IVF-PQ pipelines improve reranking throughput and latency over GPU–DRAM/SSD systems. The target is retrieval, not LLM decode.

Method

- **Store nearby:** Augment a GPU/HBM package with HBF holding full-precision vectors, avoiding host-memory round trips during reranking.
- **Search in stages:** Use IVF-PQ to identify candidates, then a near-storage search unit to support exact-distance reranking close to vector data.
- **Evaluate limits:** Model NAND subarrays, power-constrained HBF bandwidth and the retrieval pipeline to quantify data-movement and search benefits.

What Remains Open in HBM/HBF Co-Design?

Architecture trade-offs

Under a common power and package budget, when do HBM-only, hybrid HBM/HBF and full-HBF designs offer the best capacity, bandwidth and serving efficiency?

Power and thermal interaction

How do static/dynamic power, throttling, retention maintenance and extra flash traffic change sustainable bandwidth and energy per request?

Runtime and data placement

Which tiers suit weights, active KV, paused KV, reusable prefixes and MoE experts? When do migration costs outweigh reuse benefits?

Simulation fidelity

Which conclusions survive changes in hardware assumptions, latency-injection semantics and thermal models without physical HBF validation?

Candidate open questions from this review; no claim of first discovery.

Proposed Research Directions and Next Steps

Power- and Thermal-Aware HBM/HBF Co-Design for Energy-Efficient LLM Serving

1. Power-constrained architecture co-design

Compare stack/capacity allocations in HBM-only, hybrid and full-HBF designs under common package limits. Minimize energy subject to serving SLOs.

2. Thermal- and endurance-aware KV placement

Separate active, paused and reusable KV. Compare adaptive and static placement for energy, temperature and flash writes.

3. Simulator-based evaluation

Reproduce both HBFSim baselines. Calibrate power/thermal models where justified and test sensitivity to uncertain hardware assumptions.

Next steps

Read 1–2 papers/week; reproduce baselines; test one hypothesis/iteration; refine scope with the advisor.

Candidate directions; novelty and thesis scope remain to be established.